Inverse normal / percentiles

1 RECOGNISE IT

- A cumulative percentage/probability is given and the unknown is x.
- Wording: percentile, cutoff, bottom q%, top q%, central C%.
- This is area -> value, not value -> area.

2 DECISION / METHOD

1. Convert wording to cumulative-left area.
2. Get z or x with an inverse function.
3. If using z, back-transform to X.
4. Check tail and sign.

3 FORMULA

```
z_q = Φ^(-1)(q)
x_q = μ + z_q*σ
top q% uses left area 1-q
```

4 FORMULA SHEET VS WHAT I KNOW

Gives: no inverse-normal formula/function.
Still know: convert all wording to left area and back-transform from z.

5 EXCEL ROUTE

- Direct: `=NORM.INV(q,mu,sigma)`
- Via z: `=NORM.S.INV(q)`, then μ+z*σ.

6 TINY WORKED EXAMPLE

For $X \sim N(100,15^2)$, the 90th percentile is `NORM.INV(0.90,100,15)=119.22`.

7 TOP EXAM TRAPS

- Using 0.10 for the top 10% cutoff instead of 0.90.
- Using NORM.S.DIST instead of NORM.S.INV.
- Reporting z when x units are requested.
- NORM.INV asks for σ, not σ^2.

8 SAY ALOUD

Inverse normal starts with a left-tail area and returns the boundary value.

9 EXIT CHECK

What cumulative-left area represents the cutoff for the highest 5%?

ONE-GLANCE VISUAL ROUTE

